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Nonlinear Regression Adjustments of Multiple Continuous Monitoring Methods Produce Effective Characterization of Short-Term Fine Particulate Matter

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ABSTRACT

This study comprehensively characterizes hourly fine particulate matter ($PM_{2.5}$) concentrations measured via a tapered element oscillating microbalance (TEOM), β-gauge, and nephelometer from four different monitoring sites in U.S. Environment Protection Agency (EPA) Region 5 (in U.S. states Illinois, Michigan, and Wisconsin) and compares them to the Federal Reference Method (FRM). Hourly characterization uses time series and autocorrelation. Hourly data are compared with FRM by averaging across 24-hr sampling periods and modeling against respective daily FRM concentrations. Modeling uses traditional two-variable linear least-squares regression as well as innovative nonlinear regression involving additional meteorological variables such as temperature and humidity. The TEOM shows a relationship with season and temperature, linear correlation as low as 0.7924 and nonlinear model correlation as high as 0.9370 when modeled with temperature. The β-gauge shows no relationship with season or meteorological variables. It exhibits a linear correlation as low as 0.8505 with the FRM and a nonlinear model correlation as high as 0.9339 when modeled with humidity. The nephelometer shows no relationship with season or temperature but a strong relationship with humidity is observed. A linear correlation as low as 0.3050 and a nonlinear model correlation as high as 0.9508 is observed when modeled with humidity. Nonlinear models have higher correlation than linear models applied to the same dataset. This correlation difference is not always substantial, which may introduce a tradeoff between simplicity of model and degree of statistical association. This project shows that continuous monitor technology produces valid $PM_{2.5}$ characterization, with at least partial accounting for variations in concentration from gravimetric reference monitors once appropriate nonlinear adjustments are applied. Although only one regression technically meets new EPA National Ambient Air Quality Standards (NAAQS) Federal Equivalent Method (FEM) correlation coefficient criteria, several others are extremely close, showing optimistic potential for use of this nonlinear adjustment model in garnering EPA NAAQS FEM approval for continuous $PM_{2.5}$ sampling methods.

IMPLICATIONS

$PM_{2.5}$ is monitored around the world for many reasons, including legal compliance, air quality program evaluation, health effects research, and emission source identification. Once verified against FRM, continuous monitoring of $PM_{2.5}$ can supply valid, real-time air quality data, provide new information on short-term exposures for health-related studies, and be an approved method of determining EPA NAAQS compliance. Armed with information about the precision and behavior of continuous technologies in differing environmental conditions and locations, policy-makers can better tailor air quality decisions to the needs of local and global communities.

INTRODUCTION

Short-term fine continuous particulate matter ($PM_{2.5}$) measurement has recently been targeted in the field of ambient air monitoring as a feasible means of data collection. There are many benefits offered by hourly averages obtained from continuous methodologies including exploration of properties of short-term variation in particle concentrations and the potential to replace the time and labor-intensive, 24-hr current Federal Reference Method (FRM) of air quality measurement.

Particulate matter (PM) can be measured using a variety of monitoring methods. The standard, U.S. government-approved method is the FRM. It involves simple gravimetric technology (including 24- to 48-hr filter equilibration to remove water from the sample) and reports $PM_{2.5}$ concentrations averaged over 24 hr. Drawbacks of the FRM include its cumbersome, human labor-intensive process, which is unable to give current, real-time, or short-interval $PM_{2.5}$ concentrations. Automated, continuous sampling technologies such as the tapered element oscillating microbalance (TEOM), β-gauge, and nephelometer are able to remedy these problems. However, uncertainties remain as to the nature of the short-term measurements continuous technologies produce and the correlation of hourly data to FRM data.

To obtain short sampling averages, continuous monitoring technology does not measure mass directly. Instead, it measures particle mass surrogates, which are then converted into mass concentrations. The TEOM measures the oscillation of a hollow tapered element whose frequency can be proportional to mass. The TEOM operates at a constant temperature (usually 30 or 50 °C) so that the

vibrating frequency is not affected by thermal expansion or contraction caused by changing ambient temperature.[1–7] However, heating the sample, particularly on cold days, is likely to result in mass loss via volatilization. The extent of mass loss may also be related to particle composition. The β-gauge relates the attenuation of beta radiation through a filter to mass. β-gauges can be operated at ambient conditions because temperature does not affect attenuation.[8–10] Some β-gauge instruments condition the sample air to remain between 20 and 45% relative humidity (RH) but no effect of humidity on beta attenuation has been found.[11] Although it was not designed to measure a particle mass surrogate, the nephelometer uses several wavelengths of visible or infrared (IR) light to characterize light scatter from an ambient sample. It can also be operated at ambient conditions, but the use of its results as a mass surrogate is compromised by high humidity because of the extreme effect of water on aerosol light-scattering properties.[12]

Several studies have been performed examining the relationship between continuous and collocated gravimetric PM concentrations. Thirteen studies were reviewed, consisting of a total of 53 individual FRM comparison datasets: 44 TEOM,[1–7,13,14] 6 β-gauge[8–10,13] and 3 nephelometer.[12–14] Of the 51 datasets for which a squared correlation coefficient was computed, only 8 are below 0.70. Of the 39 datasets for which a basic linear model was constructed, only 4 slopes are greater than one, indicating that, in many cases, continuous monitors report lower concentration values than concurrent gravimetric monitors. Previous studies have shown TEOM to consistently underestimate PM concentrations, especially at higher inlet temperatures and during the winter season (likely because of volatization of heated mass), whereas the β-gauge provides relatively unbiased estimates, and the nephelometer has a limited history of overestimation (likely due to high scattering for wet samples).

The purpose of this project is to comprehensively explore the properties of short-term $PM_{2.5}$ measurement as compared with FRM at four monitoring sites in U.S. Environmental Protection Agency (EPA) Region 5 (in states Illinois, Michigan, and Wisconsin) for three different continuous methods. FRM-like concentrations are approximated from 24-hr averaged continuous concentrations (reported hourly) using a traditional linear model and an innovative nonlinear model expanded to include $PM_{2.5}$ concentration corrections for multiple continuous sampling methods as a function of additional ambient variables.

METHODOLOGY

This study is designed to represent air quality in Region 5 across three different methods of continuous $PM_{2.5}$ measurement. Data were compiled from four EPA monitoring locations that collected hourly $PM_{2.5}$ concentrations from at least one continuous method, as well as daily $PM_{2.5}$ concentrations from concurrent FRM. The site locations include: Mayville, WI (rural setting, agricultural land use); Grand Rapids, MI (urban setting, industrial land use); Chicago, IL (suburban setting, residential land use); and East St. Louis, IL (suburban setting, industrial land use). The type and model of each continuous instrument are: Rupprecht and Patashnick model 1400ab TEOMs operated at 50 °C, Met One model beta attenuation monitor 1020 β-gauges, and an OPTEC model NGN-2 nephelometer. The β-gauges are operated using a Smart Heater, which maintains a humidity range of 20–45% RH and limits the temperature rise of the sample filter to no more 5 °C above the temperature of the ambient air surrounding the sampler.[15] The nephelometer did not heat the sample to correct for elevated humidity. Concurrent meteorological information was collected, where available. Data were obtained either directly from the monitoring site or from EPA's online Aerometric Information Retrieval System (AIRS). A summary of data collected (not including those used for quality control) is located in Table 1.

Datasets consist of at least 1 yr of monitoring results to take into account annual variation in variables. Previous studies[6] show little difference in regression fit between analyses using complete-year versus incomplete-year data increments. Therefore, in this study, regressions use all available data for thoroughness and greatest descriptive power, with no expected bias, even though, in

Table 1. Dataset summary.

Site	Parameter	Unit	Number of Observations	Observation Type	Start Date	End Date
Mayville, WI	FRM	$\mu g/m^3$	332	24 h	01/01/2000	09/29/2002
	TEOM	$\mu g/m^3$	22,051	1 h	03/20/2000	10/31/2002
	Nephelometer	Mm^{-1}	14,879	1 h	10/01/2000	09/30/2002
	Temperature	°C	16,567	1 h	11/01/2000	10/31/2002
	Humidity	%	15,942	1 h	10/01/2000	09/30/2002
Grand Rapids, MI	FRM	$\mu g/m^3$	971	24 h	01/01/2000	09/30/2002
	TEOM	$\mu g/m^3$	20,006	1 h	01/01/2000	10/21/2002
	Temperature	°C	24,814	1 h	01/01/2000	10/31/2002
Chicago, IL	FRM	$\mu g/m^3$	119	24 h	01/01/2002	12/31/2002
	β-gauge	$\mu g/m^3$	7,489	1 h	01/01/2002	12/31/2002
	Temperature	°C	8,397	1 h	01/01/2002	12/31/2002
	Humidity	%	8,400	1 h	01/01/2002	12/31/2002
East St. Louis, IL	FRM	$\mu g/m^3$	113	24 h	01/03/2002	12/31/2002
	β-gauge	$\mu g/m^3$	7,017	1 h	01/03/2002	12/31/2002

some cases, data are not in full-year increments. Additionally, although dataset length and time collected differed between sites, individual method comparisons only used data collected over the same time frame (2002 for both sites of β-gauge and 2000–2002 for both sites of TEOM). Therefore, potential differences in interannual meteorology did not affect regression comparisons within each continuous method analysis. An investigation of potential interannual bias on PM method comparison is outside the scope of this study. Further study may also entail model verification with longer datasets, perhaps elucidating an optimal sample size from which to build FRM-predicting models.

Once data were accrued, meteorological variables and FRM measures were quality-checked for precision against an independent measure of the same variable, where available (for six of the nine noncontinuous measures datasets). The data for which no separate measure was available (Mayville humidity, Chicago and East St. Louis FRM, and the five continuous measures) were analyzed as is. Descriptive statistics, frequency distributions, and time series were calculated for each variable to pinpoint extreme values and compare PM concentrations, temperature, and humidity to previously documented patterns to help assure data validity. Autocorrelation (lagged correlation of a variable with itself) of all 14 datasets was analyzed to determine strength of seasonal patterns and influence of extreme values.

The quality-controlled data were then evaluated for associations between continuous and FRM measures of $PM_{2.5}$ using linear and nonlinear models. Data modeling compared hourly concentrations averaged over 24-hr to concurrent daily FRM concentrations. An EPA-recommended 75% completion criterion[16] was used in averaging hourly data. FRM measurements were only available for every third day for Mayville, Chicago, and East St. Louis datasets. Consequently, only one-third of the averaged hourly data was used in modeling for those locations.

The linear model least-squares regression is in the form:

$$\hat{FRM}_{\text{linear}} = \hat{b} + \hat{m}*CONT \qquad (1)$$

where $\hat{FRM}_{\text{linear}}$ is the model-estimated FRM value, *CONT* is the independent continuous method value, $\hat{b}$ is the model-estimated intercept, and $\hat{m}$ is the model-estimated slope.

The correlation coefficient derived from this regression line is a measure of the strength of linear association between FRM and continuous measurements. It indicates how well FRM can be predicted from only the value of an averaged continuous measurement.

Non-FRM measurements can be adjusted to increase correlation to FRM by using two linear models per dataset, separately fit to the data above and below predefined seasonal or meteorological cut points. Instead, the nonlinear model developed by Rizzo et al.[6] systematically identifies a cut point from the data itself as part of the regression. It does this by calculating a "knot" in the data at a particular temperature (or humidity) at which the linear relationship between continuous and FRM values changes. The model is nonlinear because the knot is not predetermined, but, instead, allowed to move depending on the particular dataset. In this way, the nonlinear model accounts for PM-affecting weather differences between sampling sites by building meteorologically comprehensive models specific to each dataset location.

The nonlinear model is constructed using nonlinear regression in SAS (originally Statistical Analysis System)[16] to fit FRM, continuous, and meteorological variables to the following relationship:

$$\begin{aligned} \hat{FRM}_{\text{nonlinear}} &= \hat{b}0 + \hat{b}1*(MET - \hat{knot}) + \hat{b}3*CONT \\ &\quad + \hat{b}4*(MET - \hat{knot})*CONT \\ &\quad (for\ MET \le \hat{knot}) \\ &\quad OR \\ &= \hat{b}0 + \hat{b}2*(MET - \hat{knot}) + \hat{b}3*CONT \\ &\quad + \hat{b}5*(MET - \hat{knot})*CONT \\ &\quad (for\ MET > \hat{knot}) \end{aligned} \qquad (2)$$

where $\hat{FRM}_{\text{nonlinear}}$ is the model-estimated FRM value, *CONT* is the independent continuous method value, *MET* is the independent meteorological value: temperature (or humidity), $\hat{b}0$ is the model-estimated intercept, $\hat{b}1$ is the model-estimated coefficient for the meteorological term for meteorological values less than the knot, $\hat{b}2$ is the model-estimated coefficient for the meteorological term for meteorological values greater than the knot, $\hat{b}3$ is the model-estimated coefficient for the continuous measurement, $\hat{b}4$ is the model-estimated coefficient of the interaction between the continuous and meteorological measurements for meteorological values less than the knot, $\hat{b}5$ is the model-estimated coefficient of the interaction between the continuous and meteorological measurements for meteorological values greater than the knot, and $\hat{knot}$ is the model-estimated meteorological value at which the linear relationship between the FRM and continuous measurement changes.

Rizzo et al. found the model to predict FRM values well from continuous TEOM measurements and temperature readings. This study further explores and develops this nonlinear model by expanding the application of the model to new datasets, monitoring methods, and a new meteorological parameter (humidity).

Once a dataset is fit to the nonlinear equation and the seven parameter estimates are obtained, predicted FRM ($\hat{FRM}_{\text{nonlinear}}$) is calculated from the nonlinear regression equation using the parameter estimates and *CONT* and *MET* values. Actual FRM is then graphed against predicted FRM and a correlation coefficient is calculated. These correlation coefficients are comparable to correlation coefficients calculated from the linear model described above. (Correlation coefficients directly from the nonlinear fit are not used in this analysis, because the introduction of more variables by definition increases correlation.) If high

correlation does not result, data are filtered according to specific omission criteria and remodeled in an effort to obtain better correlation.

RESULTS AND DISCUSSION

Quality-control checks using independent measures of the same variable result in correlation coefficients ranging from 0.8040 to 0.9998. The lowest correlation is the one comparing two measures of hourly Chicago-area humidity, likely due to the relative imprecision of humidity measurements. All other quality-control check r-values are above 0.9134, indicating a high level of precision among separate measures of the same parameter. No data were removed, with the exception of the East St. Louis point from July 4, discussed below.

Frequency distributions of $PM_{2.5}$ measurements follow lognormal distributions (determined by probability plotting). Distribution type is the same for 24- and 1-hr data. TEOM means for both hourly and 24-hr averaged data are less than concurrent FRM means for both Mayville and Grand Rapids datasets, whereas β-gauge means are greater than FRM for both Chicago and East St. Louis datasets. Hourly nephelometer measurements are collected in light-scattering units (scattering coefficients) and not mass-per-volume concentration units and are, therefore, not directly comparable to concentration. Temperature measures exhibit mixed distributions, which seem to be bimodal. RH distributions are normal. Mayville hourly humidity values include 1037 (of 15,942) observations over 100% RH, which are physically impossible measurements but were not removed for completeness.

Time series of all hourly $PM_{2.5}$ measurements show diurnal variation not discernable in daily time series. A difference between monitoring methods is observed in annual time series. FRM, β-gauge, and nephelometer time series show no evident seasonal pattern, although nephelometer data exhibit sharper, shorter-duration concentration reading peaks. However, 24-hr averaged TEOM data exhibit distinct summer peaks, in contrast to concurrent FRM.

Within time-series autocorrelation analysis shows that hourly concentrations are much more dependent on each other than daily ones. A positive PM autocorrelation greater than 2 standard errors is visible up to 3 day lags in all datasets except the Mayville nephelometer. East St. Louis β-gauge and Mayville nephelometer autocorrelations are lower, dropping below 2 standard errors within 2–4 day lags, whereas Mayville TEOM and Chicago β-gauge autocorrelations are higher, remaining above 2 standard errors for 5 day lags. The East St. Louis data may be influenced by several hours of extremely high $PM_{2.5}$ concentrations not correlated to the concentrations temporally surrounding them on July 4. The sharp spikes of extreme values observed in the nephelometer time series may decrease Mayville nephelometer autocorrelation.

Concurrent Grand Rapids TEOM data are slightly more autocorrelated than other instrument data, with positive autocorrelations above 2 standard errors for 7–9 day lags, in addition to 1–5 day lags. This fact may be indicative of TEOM seasonal variation, if TEOM concentrations are related to annually cyclic ambient temperature (because lower mass may be reported for colder weather samples, which may volatize more due to 50 °C inlet heating). Time series and autocorrelation plots are omitted for brevity.

The goal of the analysis is to develop a model for each continuous $PM_{2.5}$ measurement method into which 24-hr averaged continuous $PM_{2.5}$ data can be entered and converted into values that an FRM would have measured (i.e., "predicted" FRM). These models are developed by first quantifying, to the highest degree of correlation, the relationship between paired FRM and continuous values. The main indicator of model success is the correlation coefficient (r). (The closer r is to 1, the more successful the model. Model success was not judged within the context of statutory data quality objectives.) Other model parameter estimates refine understanding of FRM prediction capability. A summary of all correlation coefficients, sample sizes (*n*), and corresponding predictive modeling parameter estimates can be found in Tables 2 and 3.

TEOM linear model correlations are fairly high (0.7924 and 0.8484 at Mayville and Grand Rapids, respectively). A large subset of points forms a visibly linear relationship (Figure 1, solid line is regression line and dotted line is a 1-to-1 ratio line for comparison). For these points, the TEOM approximates the FRM well. However, there is a subset of points clustered above the regression line, representing data in which the TEOM underestimates the FRM (possibly because of TEOM cold weather mass volatilization, as supported by high autocorrelation). This same pattern is evident in Mayville TEOM data. When temperature is included in the TEOM models, correlation increases to 0.9192 and 0.9370 at Mayville and Grand Rapids, respectively. When humidity is included in the Mayville nonlinear model, correlation increases only slightly to 0.8102. The Grand Rapids nonlinear model (Figure 2, solid line is regression line which overlaps the 1-to-1 ratio line) adjusts the subset of TEOM-underestimated points visible in the linear model. FRM and TEOM data have different distributions across ambient temperature. All high concentration days measured by the FRM are distributed evenly across temperature (Figure 3) but TEOM-measured high concentrations are clearly biased toward higher temperatures (Figure 4), supporting the idea that mass at low temperatures tends to volatize in the heated inlet. In contrast, FRM and TEOM high concentration data are distributed evenly across ambient humidity. High TEOM autocorrelations confirm that TEOM measures vary seasonally, as does temperature but not humidity.

β-gauge linear models do not exhibit an obvious subset of nonlinear points (Figure 5, solid line is regression line and dotted line is a 1-to-1 ratio line for comparison, $r = 0.8542$). Correlation does not improve much with addition of temperature and humidity to the models ($r = 0.8628$, $r = 0.8582$, respectively). Lack of a β-gauge relationship with seasonal temperature is corroborated by low autocorrelation relative to Grand Rapids TEOM data. East St. Louis data are modeled with and without the extremely high July 4 value to ensure that one point was

Table 2. TEOM and β-gauge model parameters.

Parameter	Mayville TEOM	Grand Rapids TEOM	Chicago β-Gauge	East St. Louis β-Gauge	East St. Louis β-Gauge without July 4
Linear					
r	0.7924	0.8484	0.8542	0.9243	0.8505
n	296	774	101	93	92
m	0.9751	1.1460	0.7062	0.7816	0.6951
b	3.6919	2.5828	3.5308	1.0304	2.6593
Temperature nonlinear					
r	0.9192	0.9370	0.8628	0.9339	0.8666
n	213	774	98	89	88
*b*0	1.9352	−2.0202	3.7442	3.4993	2.4506
*b*1	0.2119	0.2153	0.1510	0.2387	0.1276
*b*2	−0.1098	0.4243	−0.0852	−0.2279	−0.0113
*b*3	1.0951	1.3922	0.6508	0.6111	0.6777
*b*4	−0.1073	−0.0859	−0.0226	−0.0313	−0.0243
*b*5	−0.0041	−0.0342	0.0060	0.0109	0.0002
knot	11.0753	16.7194	7.0833	6.8301	6.4583
Humidity nonlinear					
r	0.8102	N/A	0.8582	0.9284	0.8528
n	214	N/A	98	89	88
*b*0	3.6075	N/A	3.6733	0.1836	3.1127
*b*1	0.1149	N/A	0.1266	−0.1612	0.0395
*b*2	0.0725	N/A	0.1002	0.1254	−0.1600
*b*3	0.8460	N/A	0.7030	0.8238	0.6819
*b*4	−0.0100	N/A	−0.0046	0.0092	−0.0009
*b*5	0.0125	N/A	−0.0011	−0.0062	0.0057
knot	69.0272	N/A	73.7593	69.226	80.5956

Notes: N/A = not applicable.

Table 3. Nephelometer model parameters.

Parameter	Mayville Nephelometer	Mayville Nephelometer 24-hr RH > 90	Mayville Nephelometer 1-hr RH > 100	Mayville Nephelometer 1-hr RH > 95	Mayville Nephelometer 1-hr RH > 90
Linear					
r	0.3050	0.6068	0.5043	0.8760	0.9145
n	208	174	156	115	82
m	0.0063	0.0276	0.0240	0.1106	0.1449
b	9.9446	7.9017	8.2292	3.6867	2.8389
Temperature nonlinear					
r	0.5376	N/A	N/A	N/A	N/A
n	198	N/A	N/A	N/A	N/A
*b*0	8.2912	N/A	N/A	N/A	N/A
*b*1	−0.0397	N/A	N/A	N/A	N/A
*b*2	0.6362	N/A	N/A	N/A	N/A
*b*3	0.0000	N/A	N/A	N/A	N/A
*b*4	−0.0004	N/A	N/A	N/A	N/A
*b*5	0.0054	N/A	N/A	N/A	N/A
knot	18.8238	N/A	N/A	N/A	N/A
Humidity nonlinear					
r	0.5952	0.6123	0.8389	0.9485	0.9508
n	206	174	156	115	82
*b*0	8.0706	7.5735	2.6131	0.7673	0.6040
*b*1	0.0555	0.0232	0.1641	−0.0701	−0.0673
*b*2	0.0651	0.0848	0.1782	0.2502	1.6043
*b*3	0.0322	0.0379	0.1230	0.1397	0.1465
*b*4	−0.0005	0.0001	−0.0115	−0.0019	−0.0020
*b*5	−0.0014	−0.0016	−0.0043	−0.0093	−0.0352
knot	78.9282	78.5587	67.6983	80.3871	81.5883

Notes: N/A = not applicable.

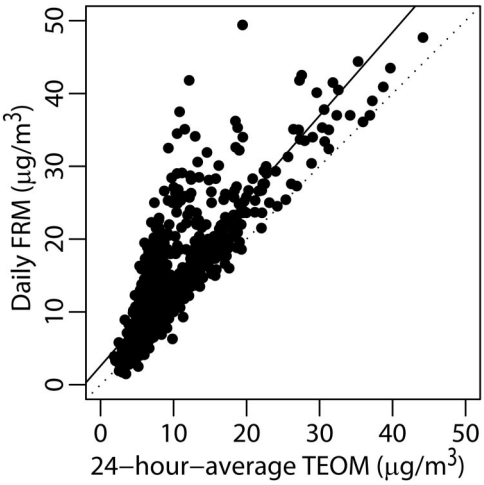


Figure 1. Grand Rapids TEOM linear model.

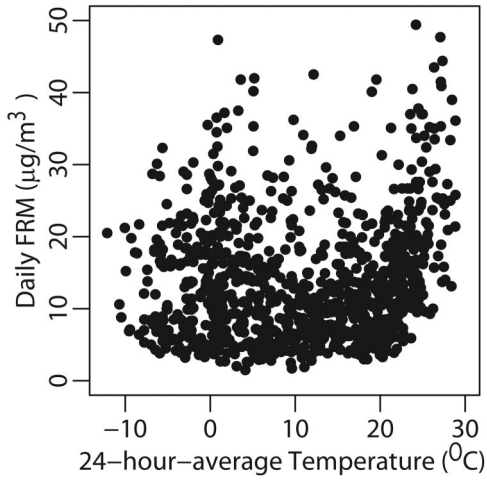


Figure 3. Grand Rapids FRM across temperature.

not imposing correlation on the sample. Without the July 4 point, East St. Louis linear correlation (r = 0.8505) is still relatively high, albeit less than correlation including the July 4 point (r = 0.9243). Unlike TEOM data, concurrent FRM and β-gauge data exhibit similar patterns across both temperature and humidity.

In contrast to TEOM and β-gauge data, nephelometer data are very difficult to correlate with collocated FRM data. Nephelometers have traditionally been used to measure visibility and the quantitative relationship to FRM data has not been well developed. One study in North Carolina found a linear relationship between nephelometer measurements and mass data.[14] However, the pattern of nephelometer to FRM in our data appears not very different from random (Figure 6, solid line is regression line and no 1-to-1 ratio line because *x* and *y* units differ), with a low correlation of 0.3050. Inclusion of humidity in the model increases correlation more than inclusion of temperature (r = 0.5952 vs. 0.5376, respectively) but correlation is still not high. These models clearly show that the nephelometer-measured aerosol light scattering is different than particle mass equivalents without further processing. Compared with Mayville FRM concentrations (equilibrated to remove water) relative to humidity (Figure 7), high Mayville nephelometer scattering coefficients are clustered around high humidities (Figure 8). Nephelometer scattering coefficients did not, however, vary across temperature. A relationship with seasonal temperature was not expected, given low nephelometer data autocorrelation.

Therefore, because nephelometer scattering coefficient is a strong nonlinear function of RH (as shown in Figure 8), and light scattering is known to significantly increase with elevated RH, data are censored at high humidity to look for strong linear PM light-scatter relationships. Observations are eliminated in subsequent models if they fit the following criteria: days with 24-hr RH > 90%, days with at least one hour RH > 100%, days with at least 1 hr RH > 95%, and days with at least 1 hr RH > 90%. As shown in Table 3, correlations increase with the elimination of additional observations. For example, the linear model with 24-hr periods that include at least 1 hr of RH > 95% are censored (Figure 9, solid line is regression line and no 1-to-1 ratio line because *x* and *y* units differ) has a correlation of 0.8760. Linear correlation reaches as high as 0.9145 when all observations on days with at least

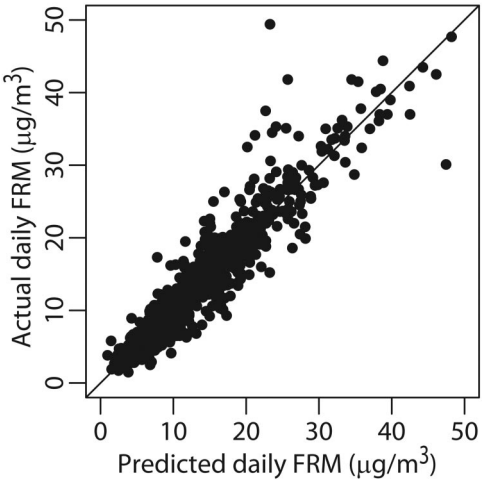


Figure 2. Grand Rapids TEOM nonlinear model with temperature.

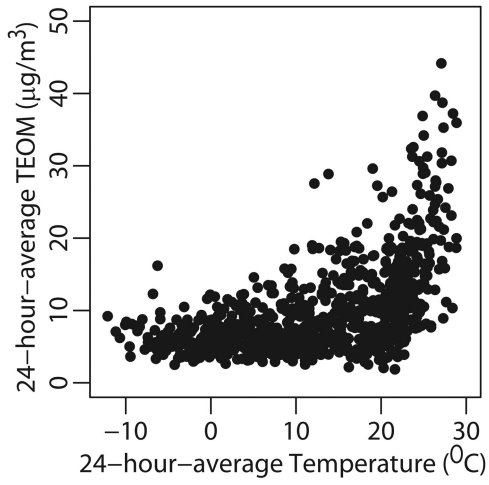


Figure 4. Grand Rapids TEOM across temperature

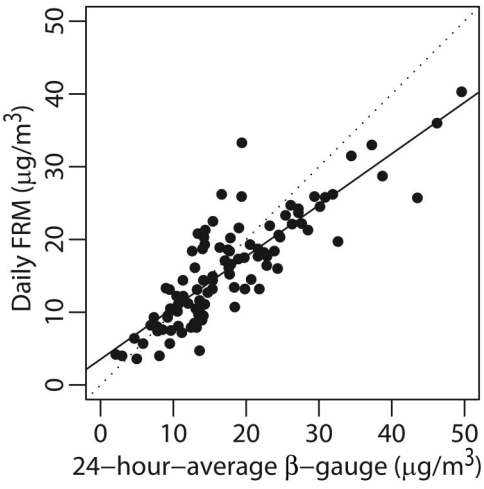


Figure 5. Chicago β-gauge linear model.

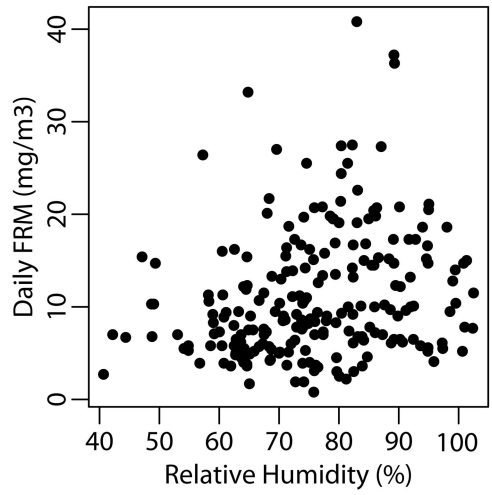


Figure 7. Mayville FRM across humidity.

1 hr of humidity over 90% are censored. Unfortunately, to achieve correlation between nephelometer and FRM data, large percentages of original data have to be cut (up to 61.23% for the model removing days with at least 1 hr RH > 90%).

It appears that the nonlinear regression approach is most effective in correcting TEOM data and less effective with β-gauge and nephelometer relationships. It also appears that the correction equation is location specific. That is, model parameters should be developed for each separate monitor. This is consistent with EPA guidance on correcting continuous FRM relationships.[17] One reason why this needs to be applied for each specific location is that the aerosol composition may be different for different locations, causing a change in the knot value or slope correction parameters. Therefore, location needs to be addressed independently. This means that to apply the nonlinear model methodology to correct, for example, TEOM data using temperature, each monitoring station must record temperature data. Luckily, this is not an issue in practice because FRM stations already log temperature data continuously. The year-to-year relationship between FRM and continuous measurements will also change if changes are made to the continuous instrument. This is a significant problem with the TEOM, which has undergone many modifications since it was released. For this nonlinear methodology to be applied successfully, it is clear that a new correction equation must be developed if an instrument is modified.

To qualify as an EPA Federal Equivalent Method (FEM) for determining compliance with $PM_{2.5}$ National Ambient Air Quality Standards (NAAQS), new criteria[17] require a linear regression correlation coefficient, r, of 0.95 or greater between candidate and concurrent reference method (if concentration coefficients of variation [CCV] ≥ 0.5). Required r decreases linearly to 0.93 as CCV decreases from 0.5 to 0.4. CCV of Mayville, Grand Rapids, and East St. Louis datasets are 0.6111, 0.6103, and 0.6019, respectively. CCV of the Chicago dataset is 0.4918, requiring a regression correlation coefficient of 0.9484 or greater. The only regression to achieve this criterion is the Mayville nephelometer nonlinear regression including humidity, filtering out all days with 1 hr of RH greater than 90% (r = 0.9508). However, several other nonlinear regressions have correlation coefficients very close to 0.95 (Grand Rapids TEOM with temperature: r = 0.9370; East

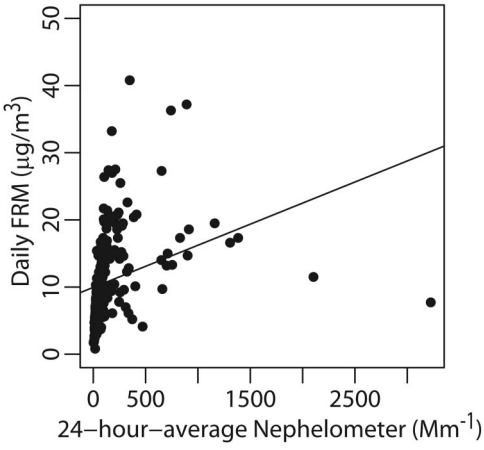


Figure 6. Mayville nephelometer linear model.

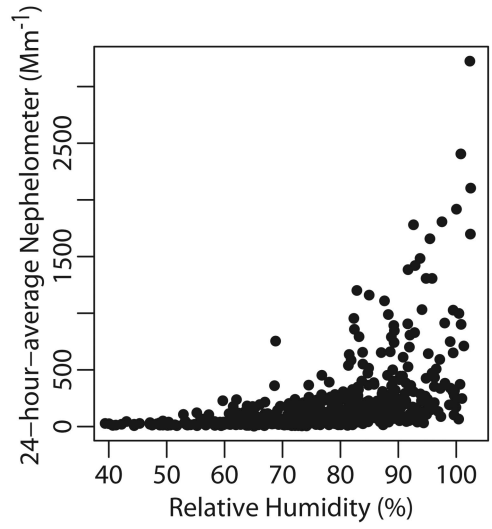


Figure 8. Mayville nephelometer across humidity.

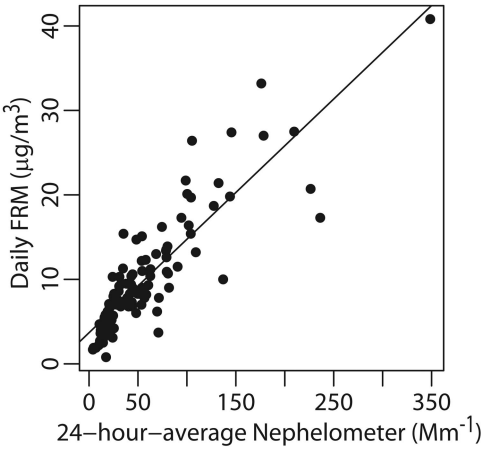


Figure 9. Mayville nephelometer linear model minus points with at least 1 hr of RH > 95%.

St. Louis β-gauge with temperature: r = 0.9339). This indicates that the nonlinear regression adjustment described in this paper, with slight further refinement, could soon be used to allow designation of $PM_{2.5}$ continuous monitoring methods as approved FEM.

CONCLUSIONS

(1) Lognormal distribution patterns of hourly $PM_{2.5}$ concentration data averaged over 24 hr are very similar to those of FRM 24-hr data. β-gauge and nephelometer time series compare well to FRM time series, whereas TEOM time series exhibit seasonal peaks related to seasonal temperature variability.

(2) In EPA Region 5, β-gauge data correlation was highest to FRM using only a basic, linear model. TEOM data correlated well to FRM after nonlinear adjustment for ambient temperature. Nephelometer data correlation to FRM improved once high humidity data points were filtered and removed and the remaining data were nonlinearly adjusted for humidity.

(3) β-gauge monitors appear unaffected by ambient temperature and humidity because data exhibit no variation of time series with season, no difference from FRM across temperature and humidity, and no major improvement in FRM correlation with nonlinear temperature and humidity models. These results show no known effect of temperature or humidity on beta attenuation technology in EPA Region 5.

(4) TEOM monitors appear to be affected by ambient temperature but not humidity because TEOM data exhibit variation of time series with season, and there is a difference from FRM across temperature but not humidity. Also, FRM to TEOM correlation improves with nonlinear temperature but not humidity models. Interaction of instrument technology with ambient temperature likely explains this effect via the heating of semivolatile compounds, which, in the colder winter samples tend to volatize more than in the warmer summer samples. EPA Region 5 winter underestimation is best accounted for when data collected at low and high temperatures were modeled separately within one model. Ambient humidity does not appear to have an effect on sample volatization, indicating that the same amount of sample volatizes in the 50 °C inlet on both wet and dry days.

(5) Nephelometer monitors appear to be affected by ambient humidity and not temperature because nephelometer data exhibit variation of time series with humidity, and there is a difference from FRM across humidity but not temperature. Correlation to FRM improves more with humidity censoring and nonlinear humidity models than with a nonlinear temperature models. These results are consistent with the known relationship between RH and light scatter. Linear PM light-scatter models improve when days with high RH are removed from the analysis. Nonlinear PM light-scatter analysis was generally not successful using temperature as an independent variable, but was successful when models were modified with humidity data.

(6) This work contributes to the growing body of knowledge concerning hourly $PM_{2.5}$ data and comparison to FRM. Once hourly $PM_{2.5}$ is better characterized for other sites and datasets, it can be used to study short-term health effects, develop short-term PM standards, and report faster, more timely air quality conditions relative to current standards. Additionally, this nonlinear regression adjustment method may make possible the designation of these studied continuous monitoring methods as approved FEM for the measurement of $PM_{2.5}$ NAAQS compliance. In the process of doing so, however, policy-makers should be aware of the behavior of continuous technologies relative to regional location and varying environmental conditions.

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